Vrushabh Desai

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Summary

Senior Software Engineer with 7+ years of experience in C/C++ systems software, embedded development, and computer vision. Built remote diagnostics tools for production vision systems, designed and shipped a cross-platform vision SDK, and developed embedded firmware and real-time control systems for electric vehicle and motorsport platforms.

Experience

Senior Software Engineer - R&D | Cognex Corporation, Natick, MA **Jun 2021 - Present**

- Architected a **C/C++ remote diagnostics** platform for embedded vision systems, enabling engineers to debug Linux-based cameras from Windows and view live graphics and algorithm state, reducing hardware issue resolution time from **4 hours to 15-30 minutes**.
- Built a **C++ diagnostic replay framework** that captured runtime system state from production devices and reproduced customer failures in desktop tools, cutting investigation time from **1 day to 1 hour**.
- Designed and shipped the **Cognex Development Kit (CDK)**, a cross-platform software development kit with C++, Python, and C# interfaces that unified internal 2D, 3D, and AI-based vision tools into a single developer platform, reducing on-device tool deployment time by **60%** and enabling teams to launch new tools faster.
- Drove **CDK adoption** across different cognex embedded products by partnering with customers and downstream engineering teams to resolve deployment constraints and production integration issues.
- Eliminated a **performance bottleneck** in a 3D vision point-cloud pipeline by replacing deep-copy ownership with pointer-based semantics, improving benchmark performance by **2x**.
- Owned CDK continuous integration and delivery (CI/CD) architecture in TeamCity, introducing **infrastructure-as-code** pipelines and on-demand build agents that **reduced build times by 4x** across ARM and x64 platforms.
- Led modernization planning for a legacy **Cognex Vision Library (CVL)** supporting a $10M+ revenue product by defining MVP scope, aligning stakeholders, and prioritizing engineering investments to accelerate delivery and lower program risk.
- Served as **Scrum Master** and **Release Shepherd** for CDK, coordinated cross-functional CDK releases across Windows installers, Debian packages, Python wheels, and NuGet packages, improving release readiness and execution across multiple distribution channels.
- Built **AI-assisted** release and dependency-management workflows that automated build validation, release-note generation, and pull-request creation, cutting release effort by **60%**.

Embedded Systems Engineer | Jindal Mobilitrics, Mumbai **Jun 2018 - Jun 2019**

- Led controls development for an autonomous delivery robot, building Raspberry Pi-based perception and control software with OpenCV and YOLOv3 and running system-level testing to validate lane-following and obstacle-avoidance behavior on embedded hardware.
- Architected a custom microcontroller-based Vehicle Control Unit (VCU) for an electric motorbike, developed embedded C firmware and CAN bus communication software to integrate sensor inputs into a real-time vehicle control system.
- Designed a 3 kW battery pack, Battery Management System (BMS), and low-voltage electrical harness for an electric motorbike, contributing end-to-end power-system architecture and vehicle-electronics integration alongside embedded control development.

Technical Projects

DJS Racing Formula Student Team | Vehicle Systems & Telemetry Lead **Jun 2016 - May 2018**

- Architected the race car's vehicle electronics and telemetry stack, integrating the engine control unit (ECU), power distribution, and real-time data acquisition systems, and developed a closed-loop drag reduction system (DRS) controller for track-based aerodynamic control.
- Built real-time embedded control systems for a Formula Student race car, including paddle-actuated electronic shifting and drivetrain control, improving response time and reliability under high-speed racing conditions.
- Delivered vehicle systems for an **award-winning Formula Student race car** that won Best Designed Car at Formula Bharat in 2017 and 2018, placed as the second-best Asian team at Formula Student Germany 2017, and won the Cost event at Formula Student Austria 2018.

Skills

Programming Languages: C++, C, Embedded C, Python, C#, Bash, Kotlin

Build & Toolchain: CMake, Conan, Ninja, GCC, Clang, MSVC, Visual Studio

CI/CD & DevOps: TeamCity, Jenkins, Docker, AWS, JFrog Artifactory, Bitbucket Server, Git

Testing & Quality: BullseyeCoverage, CTest, pytest, Google Benchmark, clang-tidy, AddressSanitizer

Embedded, Vision & Robotics Tools: CAN bus, Raspberry Pi, OpenCV, YOLOv3, ROS2, Gazebo, MATLAB, Atmel Studio

Release & Packaging: Debian, Python wheels, NuGet, NSIS, InstallShield

Education

MS in Robotics Engineering | Worcester Polytechnic Institute, Worcester, MA | **CGPA: 4.00** **Aug 2019 - May 2021**

BE in Electronics Engineering | D. J. Sanghvi College of Engineering, Mumbai, India | **CGPA: 3.7** **Aug 2014 - Jun 2018**